First Non Repeating Character

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The problem

- Index of the first lowercase char occurring exactly once, else -1.

Intuition

- Two passes: first count every character, then find the first with a count of one.

Approach

- 1. Count frequencies in a map
- 2. Scan the string again
- 3. Return the first index with count one, else minus one

Java solution

```java
public class FirstNonRepeatingCharacter {
    public static int firstNonRepeatingCharacter(String s) {
        int[] freq = new int[26];
        for (char c : s.toCharArray()) freq[c - 'a']++;
        for (int i = 0; i < s.length(); i++) if (freq[s.charAt(i) - 'a'] == 1) return i;
        return -1;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1) (26 counters)

Recap

- Two passes: first count every character, then find the first with a count of one.
- Complexity: O(n)